

Unit 27: The Fundamental Theorem of Calculus

1. The idea, in plain words

Here's the payoff for everything you've built so far. Back in Units 24–25, computing an exact area meant grinding through a limit of Riemann sums — real work, every single time. The Fundamental Theorem of Calculus (FTC) says you never have to do that again. It reveals a stunning, tight connection between the two big ideas of calculus — derivatives and areas — and turns definite integral computation into something you can do in a few lines.

The Fundamental Theorem has two parts, and they say two different (but related) things.

Part 1 — differentiating an integral gives you back the original function. If you build a new function by integrating f from a fixed starting point up to a variable endpoint x , then differentiating that new function just hands you $f(x)$ right back. Picture it like this: integrating is “accumulating,” and differentiating measures “the rate of accumulation” — and the rate at which you're accumulating f is, quite naturally, just f itself. Integration and differentiation are, in a real sense, opposite operations that undo each other.

Part 2 — the one you'll use constantly — lets you evaluate any definite integral using an antiderivative. If F is any antiderivative of f (found using everything you learned in Unit 23), then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

That's it. No limits, no rectangles — just find an antiderivative, plug in the top number, plug in the bottom number, and subtract. This single fact is why antiderivatives matter so much: they're not just an abstract “undo the derivative” exercise — they're the literal key that unlocks exact area computation.

2. Toolbox

FTC Part 1 (the derivative of an accumulation function):

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

FTC Part 1, chain rule version (when the upper limit is itself a function of x , not just plain x):

$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

FTC Part 2 (the Evaluation Theorem) — the one you'll use most often:

$$\int_a^b f(x) dx = F(b) - F(a) = \left[F(x) \right]_a^b$$

where F is any antiderivative of f (found using the techniques from Unit 23). Notice: since the constant $+C$ would appear in both $F(b)$ and $F(a)$, it always cancels out during the subtraction — so you can safely ignore it entirely for definite integrals.

The practical procedure for Part 2:

1. Find an antiderivative $F(x)$ of $f(x)$ (no need for $+C$).
2. Evaluate F at the top limit, and at the bottom limit.
3. Subtract: (value at top) $-$ (value at bottom).

3. Common mistakes

- Subtracting in the wrong order. It's always $F(b) - F(a)$ — top limit's value minus bottom limit's value — never the reverse.
- Forgetting the chain rule when the upper limit of an FTC Part 1 problem isn't plain x . If the upper limit is something like x^2 or $\sin x$, you must multiply by the derivative of that upper limit too.
- Trying to carry a $+C$ through a definite integral evaluation. It's unnecessary and just adds confusion — the constant always cancels, so leave it out entirely once you're evaluating with specific bounds.
- Algebra slips when evaluating F at each limit. Take it slow, especially with fractions or negative numbers — a small arithmetic error here throws off the whole final answer.
- Not simplifying the integrand first when it's a disguised sum (like a fraction that splits into several power-rule-friendly pieces) before hunting for the antiderivative.

4. Problem Set

Warm-up

1. $\int_0^2 x^2 dx$
2. $\int_1^3 (2x + 1) dx$
3. $\int_0^4 x dx$
4. $\int_{-1}^2 3x^2 dx$
5. $\int_0^1 (x^3 + x) dx$
6. $\int_1^2 4 dx$
7. $\int_0^\pi \sin x dx$

Standard

8. $\int_0^{\pi/2} \cos x dx$
9. $\int_1^4 \sqrt{x} dx$
10. $\int_1^2 \frac{1}{x^2} dx$

11. $\int_0^2 (x^2 - 2x + 3) dx$
12. Find $\frac{d}{dx} \left[\int_2^x (t^2 + 1) dt \right]$.
13. Find $\frac{d}{dx} \left[\int_0^x \sin t dt \right]$.
14. $\int_{-2}^2 x^3 dx$ (confirm using FTC that this matches what symmetry alone would predict)

Challenge

15. Find $\frac{d}{dx} \left[\int_1^{x^2} (t^3 + 1) dt \right]$.
16. Find $\frac{d}{dx} \left[\int_0^{\sin x} \sqrt{1+t^2} dt \right]$.
17. $\int_1^2 \frac{x^4 + x^2}{x^2} dx$ (simplify first!)
18. Evaluate $\int_0^3 |x-1| dx$ (split into pieces at $x=1$ first).
19. Given $F(x) = \int_0^x (t^2 - 4) dt$, find the critical points of $F(x)$ and classify each as a local max or local min.

Applied

20. A particle's velocity is $v(t) = 3t^2 - 6t$ m/s. Find its displacement (net change in position) over $[0, 3]$ using FTC Part 2.
21. Water flows into a tank at a rate of $r(t) = 6t + 2$ gal/min. Find the total gallons collected over $[0, 5]$ minutes using FTC Part 2.
22. A company's marginal cost is $MC(x) = 2x + 10$. Find the total additional cost of increasing production from $x = 5$ to $x = 10$ using FTC Part 2.
23. Total distance traveled is modeled by $D(t) = \int_0^t v(s) ds$, where $v(s) = 4s + 1$. Find $D'(t)$ using FTC Part 1, and explain what it represents physically.